RESONANCE LOWER BOUNDS FOR THE NUMBER-FIELD PILTZ DIVISOR ERROR WITH SPLITTING-DENSITY LOGARITHMIC EXPONENTS

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Allowed standard prerequisites (editorial).

- Dedekind zeta functions, prime splitting densities, Voronoi transforms and resonance bounds for exponential sums. Original exact divisor/error and density definitions, cited resonance input and full source-local number-field construction with complete bibliography retained.

1. Introduction

The study of error terms in various counting functions is a central theme in analytic number theory. The Dirichlet divisor problem and the Gauss circle problem are two classical examples that ask to estimate the discrepancy between an arithmetic counting function and its asymptotic approximation. The error terms of the above said problems are

$$\Delta(x) = \sum_{n\leq x} d(n) - x\log x - (2\gamma-1)x, \quad \text{and} \quad R(x) = \sum_{\substack{a^2+b^2\leq x \\ a,b\in\mathbb{Z}}} 1 - \pi x, \tag{1.1}$$

respectively, where $d(n)$ counts the number of divisors for a positive integer n and $\gamma = 0.5772\cdots$ is the Euler's constant. Both, the Gauss circle problem and the Dirichlet divisor problem seek upper bounds of the form[1] $R(x) \ll x^{\theta_1+\epsilon}$ and $\Delta(x) \ll x^{\theta_2+\epsilon}$, with θ_i as small as possible. It is conjectured that $\theta_i = 1/4$ for both the problems. The sharpest known value of θ_i, due to Huxley [5, 6], has long been $\theta_i = \frac{131}{416} = 0.3149\cdots$. However, Li and Yang [7] recently refined this result by improving the bound at the fourth decimal place, showing that $\theta = 0.314483\cdots$. It is also of interest to understand how large $R(x)$ and $\Delta(x)$ could be as $x\to\infty$. In this direction, Soundararajan [11] established [2]

$$\Delta(x) = \Omega\left((x\log x)^{1/4}(\log_2 x)^{\frac{3}{4}(2^{\frac{4}{3}-1})}(\log_3 x)^{-5/8}\right)$$

and

$$R(x) = \Omega\left((x\log x)^{1/4}(\log_2 x)^{\frac{3}{4}(2^{\frac{4}{3}-1})}(\log_3 x)^{-5/8}\right).$$

Using the resonance method, the second author [8] has improved these Ω-bounds by raising the exponent of $\log_3 x$ from $-5/8$ to $-3/8$. The aim of this article is to extend the approach in [8] in number fields, which includes the Gauss circle problem and the Dirichlet divisor problem.

Let $\mathbb{K}$ be an algebraic number field of degree $[\mathbb{K}:\mathbb{Q}] = m = r_1 + 2r_2$, where r_1 and r_2 denote the number of real and complex embeddings of $\mathbb{K}$, respectively, and $\mathcal{O}_{\mathbb{K}}$ be its ring of integers. Let D

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[1] The symbol ϵ always denotes an arbitrary small positive constant, whose value may varry as needed.

[2] Throughout this paper we denote $\log_2 X = \log\log X$ and $\log_3 X = \log\log\log X$.

be the absolute value of the discriminant of $\mathbb{K}$ and $N(\mathfrak{a})$ denotes the absolute norm of $\mathfrak{a}$. Then the Dedekind zeta function associated to $\mathbb{K}$ is defined as

$$\zeta_{\mathbb{K}}(s) = \sum_{\mathfrak{a}\subset\mathcal{O}_{\mathbb{K}}} \frac{1}{N(\mathfrak{a})^s} = \prod_{\mathfrak{p}\subset\mathcal{O}_{\mathbb{K}}} \left(1 - \frac{1}{N(\mathfrak{p})^s}\right)^{-1},$$

for all $s \in \mathbb{C}$ with $\mathrm{Re}(s) > 1$, where the sum and the product are taken over all non zero integral ideals $\mathfrak{a}$ and prime ideals $\mathfrak{p}$ of $\mathcal{O}_{\mathbb{K}}$, respectively. The function $\zeta_{\mathbb{K}}(s)$ can be continued analytically to a meromorphic function with a unique simple pole at $s = 1$. Moreover, it satisfies the functional equation $\zeta_{\mathbb{K}}(s) = \chi_{\mathbb{K}}(s)\zeta_{\mathbb{K}}(1-s)$, where

$$\chi_{\mathbb{K}}(s) = 2^{m(s-1)}\pi^{ms} D^{1/2-s} \frac{\sec^{r_1+r_2}(\pi s/2)\,\mathrm{cosec}^{r_2}(\pi s/2)}{\Gamma(s)^m}. \tag{1.2}$$

The number field analogue of k-fold divisor function is defined by

$$d_{\mathbb{K}}^{(k)}(n) = \sum_{N(\mathfrak{a}_1\mathfrak{a}_2\cdots\mathfrak{a}_k)=n} 1, \tag{1.3}$$

where the sum is taken over k-tuples of ideals in the number field $\mathbb{K}$ whose product has norm n. The arithmetic function $d_{\mathbb{K}}^{(k)}(n)$ satisfies the following identity:

$$\zeta_{\mathbb{K}}^k(s) = \sum_{n=1}^{\infty} \frac{d_{\mathbb{K}}^{(k)}(n)}{n^s}, \qquad \mathrm{Re}(s) > 1.$$

In analogy to the classical Dirichlet divisor problem in (1.1), our aim is to investigate the behavior of the error term in the asymptotic formula

$$\Delta_{\mathbb{K}}^{(k)}(x) = \sum_{n\le x} d_{\mathbb{K}}^{(k)}(n) - \operatorname*{Res}_{w=1}\left(\zeta_{\mathbb{K}}^k(w)\frac{x^w}{w}\right), \tag{1.4}$$

as $x \to \infty$. The problem of evaluating the error term $\Delta_{\mathbb{Q}}^{(k)}$ is called the *Piltz divisor problem*. There are very few results on this problem over number field. Nowak [9] showed that the upper bound estimate for $\Delta_{\mathbb{K}}^{(k)}$ is as follows:

$$\Delta_{\mathbb{K}}^{(k)}(x) \ll \begin{cases} x^{1-\frac{2}{mk}+\frac{8}{mk(5mk+2)}}(\log x)^{k-1-\frac{10(k-2)}{5m+2}}, & \text{for } 3 \le mk \le 6, \\ x^{1-\frac{2}{mk}+\frac{3}{2m^2k^2}}(\log x)^{k-1-\frac{2(k-2)}{mk}}, & \text{for } mk \ge 7, \end{cases}$$

where $m = [\mathbb{K} : \mathbb{Q}] \ge 2$. Recently, Takeda [14] improved this upper bound, showing that

$$\Delta_{\mathbb{K}}^{(k)}(x) \ll_{\epsilon} x^{(2mk-3)/(2mk+1)+\epsilon} D^{2k/(2mk+1)+\epsilon},$$

holds for every $mk \ge 4$.

For the lower bound estimate, we need to introduce some notations from algebraic number theory. Most of the Ω-estimates for $\Delta_{\mathbb{K}}^{(k)}$ depend on Dirichlet densities. For $0 \le \nu \le m$, P_ν denotes the set of all rational primes which are unramified in $\mathbb{K}$ and which split into exactly ν distinct $\mathcal{O}_{\mathbb{K}}$-prime ideals of degree 1. To define the Dirichlet density δ_ν precisely, we set $\mathbb{K}^G$ to be the Galois closure of $\mathbb{K}/\mathbb{Q}$, $G = \mathrm{Gal}\left(\mathbb{K}^G/\mathbb{Q}\right)$ its Galois group, and $H = \mathrm{Gal}\left(\mathbb{K}^G/\mathbb{K}\right)$ the subgroup of G corresponding to $\mathbb{K}$. Then the Dirichlet densities δ_ν are defined as

$$\delta_\nu = \frac{\left|\left\{\sigma \in G : \left|\left\{\psi \in G : \sigma \in \psi H \psi^{-1}\right\}\right| = \nu\,|H|\right\}\right|}{|G|}, \tag{1.5}$$

and these constants satisfy the condition $\sum_{\nu=1}^{m}\nu\delta_\nu = 1$. Let R be the cardinality of the set $\mathcal{I} = \{1 \leq \nu \leq m : \delta_\nu > 0\}$. In this setup, Szegö and Walfisz [12, 13] first derived the estimate

$$\Delta_{\mathbb{K}}^{(k)}(x) = \Omega_*\left((x\log x)^{\frac{mk-1}{2mk}}(\log_2 x)^{k-1}\right).$$

Later, Hafner [4] successfully improved this lower bound and proved

$$\Delta_{\mathbb{K}}^{(k)}(x) = \Omega_*\left((x\log x)^{\frac{mk-1}{2mk}}(\log_2 x)^{\beta'}\exp(-c_3\sqrt{\log_3 x})\right),$$

where c_3 is some positive constant and

$$\beta' = \frac{mk-1}{2mk}\left(\left(\sum_{\nu=1}^{k}\delta_\nu\,\nu\log\nu\right)k + k\log k - k + 1\right) + k - 1.$$

Here, the Ω-estimates are defined as follows:

$$\Omega_* = \begin{cases}\Omega_\pm, & \text{if } mk \geq 4 \text{ or } \mathbb{K} \text{ is cubic and not totally real,}\\ \Omega_+, & \text{if } k = 2,3 \text{ and } \mathbb{K} = \mathbb{Q}, \text{ or } k = 1, m = 2,3 \text{ and } \mathbb{K} \text{ is totally real,}\\ \Omega_-, & \text{if } k = 1 \text{ and } \mathbb{K} \text{ is quadratic imaginary.}\end{cases}$$

Using Soundararajan's [11] method, Girstmair et al. [2] refined Hafner's [4] Ω-result. Indeed, they showed that

$$\Delta_{\mathbb{K}}^{(k)}(x) = \Omega\Big((x\log x)^{\frac{mk-1}{2mk}}(\log_2 x)^{\beta}(\log_3 x)^{\gamma'}\Big), \tag{1.6}$$

with

$$\beta = \frac{mk+1}{2mk}\left(\sum_{\nu=1}^{m}\delta_\nu(k\nu)^{\frac{2mk}{mk+1}} - 1\right), \quad \gamma' = -\frac{mk+1}{4mk}R - \frac{mk-1}{2mk}. \tag{1.7}$$

Motivated by the approach in [1], the second author uses resonance method to general exponential sums and gives detailed lower bounds for these sums in [8, Theorem 5]. We use this to improve the Ω-bounds in (1.6), which yields our main result:

Theorem 1.1. *Let $\mathbb{K}$ be any arbitrary algebraic number field of degree m and k be any positive integer with $mk \geq 2$. Then, for $x \to \infty$, the error term $\Delta_{\mathbb{K}}^{(k)}(x)$ in (1.4) satisfies*

$$\Delta_{\mathbb{K}}^{(k)}(x) = \Omega\Big((x\log x)^{\frac{km-1}{2km}}(\log_2 x)^{\beta}(\log_3 x)^{\gamma}\Big), \tag{1.8}$$

where

$$\beta = \frac{mk+1}{2mk}\left(\sum_{\nu=1}^{m}\delta_\nu(k\nu)^{\frac{2mk}{mk+1}} - 1\right) \quad \text{and} \quad \gamma = -\frac{km+1}{4mk}R.$$

Moreover, if r_1 denotes the number of real conjugates of $\mathbb{K}$, in (1.8) we may replace Ω with Ω_+ when $kr_1 \equiv 3\,(\mathrm{mod}8)$ and Ω with Ω_- when $kr_1 \equiv 7\,(\mathrm{mod}8)$.

The above theorem includes (1.1) as special cases. The Gauss circle problem corresponds to $\mathbb{K} = \mathbb{Q}(i)$ and $k = 1$, while the Dirichlet divisor problem corresponds to $\mathbb{K} = \mathbb{Q}$ and $k = 2$. In Section 4, we discuss some special examples.

The structure of the paper is as follows. In Section 2, we present a result due to the second author, which serves as a key ingredient in the proof of our main theorem. We also generalize a Voronoi-type formula due to Soundararajan [11] in the context of number fields. Section 3 is devoted to the proof of Theorem 1.1, and in Section 4, we discuss some immediate examples of this theorem.

2. Preliminaries

We recall a theorem due to the second author [8], which plays a central role in establishing our main result. Let $(\lambda_n)_{n=1}^{\infty}$ be a non-decreasing sequence of positive real numbers and α be a positive real parameter. We consider a finite set $\mathcal{M} \subseteq \{\lambda_n : C_1\alpha \leq \lambda_n \leq 2\alpha\}$, where $0 < C_1 < 2$ such that the elements of $\mathcal{M}$ are linearly independent over $\mathbb{Q}$. Let M be the cardinality of the set $\mathcal{M}$. Additionally, given A_1, we consider real numbers A_2, A_3, A_4 satisfying $0 < A_4 < A_3 < A_2 < A_1$, and let θ be any real number.

Theorem 2.1. *Let $(f(n))_{n=1}^{\infty}$ be a sequence of positive real numbers. Then, for large X, we have*

$$\max_{X^{A_3}/2\leq x\leq 2A_2^2X^{A_2}\log^2 X}\left|\sum_{n\leq X^{A_1}} f(n)\cos\left(x\lambda_n+\theta\right)\right| \geq \frac{\pi}{4e}\sum_{n\in\mathcal{M}} f(n) + O\left(X^{A_3-A_2}e^{2M/C_1}\left(\sum_{\lambda_n\leq 4\alpha} f(n)\right)\right) + O\left(\frac{X^{-A_4}}{\alpha}\sum_{n\leq X^{A_1}} f(n)\right).$$

To complete the proof of our main theorem, we prove the following proposition, which is a generalization of Soundararajan's result [11] in number fields. This proposition contains a Vornoi type summation formula for the error term $\Delta_{\mathbb{K}}^{(k)}(x)$.

Proposition 2.2. *Let $\mathbb{K}$ be any number field of degree $m = r_1+2r_2$, where r_1 and r_2 denote the number of real and complex embeddings of $\mathbb{K}$, respectively. Let D be the absolute value of the discriminant of $\mathbb{K}$. Suppose $x \geq 2$ and $\alpha \geq 2$ be two real parameters. Then, for a fixed positive integer k, we have*

$$\frac{\alpha^{\frac{1}{mk}}}{\sqrt{\pi}}\int_{-\infty}^{\infty}\Delta_{\mathbb{K}}^{(k)}\left(x^{mk}e^{u/x}\right)\exp\left(-u^2\alpha^{\frac{2}{mk}}\right)\mathrm{d}u = O\left(t^{\frac{mk}{2}-\frac{3}{5}}\alpha^{\frac{2}{mk}}\right) + \frac{D^{\frac{1}{2m}}}{\pi\sqrt{mk}}x^{\frac{mk-1}{2}}\sum_{n=1}^{\infty}\frac{d_{\mathbb{K}}^{(k)}(n)}{n^{\frac{mk+1}{2mk}}}\exp\left(-\pi^2\left(\frac{n}{D^k\alpha}\right)^{\frac{2}{mk}}\right)\cos\left(2\pi mk\left(\frac{n}{D^k}\right)^{\frac{1}{mk}}x+\frac{kr_1-3}{4}\pi\right). \quad (2.1)$$

Proof. Let $D_{\mathbb{K}}^{(k)}(x) = \sum_{n\leq x} d_{\mathbb{K}}^{(k)}(n)$. We start with the following integral:

$$\frac{\alpha^{\frac{1}{mk}}}{\sqrt{\pi}}\int_{-\infty}^{\infty}D_{\mathbb{K}}^{(k)}\left(x^{mk}e^{u/x}\right)\exp\left(-u^2\alpha^{\frac{2}{mk}}\right)\mathrm{d}u.$$

Using Perron's formula, for $c > 1$, the above integral becomes

$$\begin{aligned}
&\frac{\alpha^{\frac{1}{mk}}}{\sqrt{\pi}}\int_{-\infty}^{\infty}\int_{c-i\infty}^{c+\infty}\zeta_{\mathbb{K}}(s)^k x^{mks}\exp\left(\frac{us}{x}-u^2\alpha^{\frac{2}{mk}}\right)\frac{\mathrm{d}s}{s}\mathrm{d}u\\
=&\frac{1}{2\pi i}\int_{c-i\infty}^{c+\infty}\zeta_{\mathbb{K}}(s)^k x^{mks}\frac{\alpha^{\frac{1}{mk}}}{\sqrt{\pi}}\int_{-\infty}^{\infty}\exp\left(-\left(u\alpha^{\frac{1}{mk}}-\frac{s}{2\alpha^{\frac{1}{mk}}x}\right)^2+\frac{s^2}{4\alpha^{\frac{2}{mk}}x^2}\right)\mathrm{d}u\frac{\mathrm{d}s}{s}\\
=&\frac{1}{2\pi i}\int_{c-i\infty}^{c+\infty}\zeta_{\mathbb{K}}(s)^k x^{mks}\exp\left(\frac{s^2}{4\alpha^{\frac{2}{mk}}x^2}\right)\frac{\mathrm{d}s}{s}.
\end{aligned}$$

Now we consider $d = -1/\log x$. We shift the above line of integration to the line $d - i\infty$ to $d + i\infty$. Here, we see that the integrand has poles at $s = 0$ and $s = 1$. Since,

$$\operatorname*{Res}_{s=0}\zeta_{\mathbb{K}}(s)^k\frac{x^{mks}}{s}\exp\left(\frac{s^2}{4\alpha^{\frac{2}{mk}}x^2}\right) = \zeta_{\mathbb{K}}(0)^k,$$

we get an amount $O(1)$ for the pole at $s=0$ and for the pole at $s=1$, we can write the following:

$$\operatorname*{Res}_{s=1}\zeta_{\mathbb{K}}(s)^k\frac{x^{mks}}{s}\exp\left(\frac{s^2}{4\alpha^{\frac{2}{mk}}x^2}\right)=\frac{\alpha^{\frac{1}{mk}}}{\sqrt{\pi}}\int_{-\infty}^{\infty}e^{-u^2\alpha^{\frac{2}{mk}}}\left(\operatorname*{Res}_{s=0}\zeta_{\mathbb{K}}(s)^k\frac{\left(x^{mk}e^{u/x}\right)^s}{s}\right)\mathrm{d}u.$$

Note that for any $x>0$,

$$\Delta_{\mathbb{K}}^{(k)}(x)=D_{\mathbb{K}}^{(k)}(x)-\operatorname*{Res}_{s=1}\zeta_{\mathbb{K}}(s)^k\frac{x^s}{s}.$$

By Cauchy residue theorem and the above identity, we arrive

$$\frac{\alpha^{\frac{1}{mk}}}{\sqrt{\pi}}\int_{-\infty}^{\infty}\Delta_{\mathbb{K}}^{(k)}\left(x^{mk}e^{u/s}\right)\exp\left(-u^2\alpha^{\frac{2}{mk}}\right)\mathrm{d}u=\frac{1}{2\pi i}\int_{d-i\infty}^{d+i\infty}\zeta_{\mathbb{K}}(s)^kx^{mks}\exp\left(\frac{s^2}{4\alpha^{\frac{2}{mk}}x^2}\right)\frac{\mathrm{d}s}{s}+O(1). \tag{2.2}$$

Since, $\zeta_{\mathbb{K}}(s)$ satisfies the functional equation $\zeta_{\mathbb{K}}(s)=\chi_{\mathbb{K}}(s)\zeta_{\mathbb{K}}(1-s)$, where $\chi_{\mathbb{K}}(s)$ is defined in (1.2) and $\zeta_{\mathbb{K}}(1-s)=\sum_{n=1}^{\infty}d_{\mathbb{K}}^{(k)}(n)n^{s-1}$ holds, equation (2.2) becomes

$$\frac{\alpha^{\frac{1}{mk}}}{\sqrt{\pi}}\int_{-\infty}^{\infty}\Delta_{\mathbb{K}}^{(k)}\left(x^{mk}e^{u/s}\right)\exp\left(-u^2\alpha^{\frac{2}{mk}}\right)\mathrm{d}u=\sum_{n=1}^{\infty}\frac{d_{\mathbb{K}}^{(k)}(n)}{n}\mathcal{I}_n+O(1), \tag{2.3}$$

where

$$\mathcal{I}_n=\frac{1}{2\pi i}\int_{d-i\infty}^{d+i\infty}\chi_{\mathbb{K}}(s)^k\left(x^{mk}n\right)^s\exp\left(\frac{s^2}{4\alpha^{\frac{2}{mk}}x^2}\right)\frac{\mathrm{d}s}{s}.$$

We see that

$$\frac{1}{2\pi i}\int_{d-i}^{d+i}\chi_{\mathbb{K}}(s)^k\left(x^{mk}n\right)^s\exp\left(\frac{s^2}{4\alpha^{\frac{2}{mk}}x^2}\right)\frac{\mathrm{d}s}{s}\ll n^d\log x.$$

Also, observe that the integrand of $\mathcal{I}_n$ at $d+it$ is the complex conjugate of of the integrand at $d-it$. Then

$$\mathcal{I}_n=\operatorname{Re}\frac{1}{\pi i}\int_1^{\infty}\chi_{\mathbb{K}}(d+it)^k\left(x^{mk}n\right)^{d+it}\exp\left(\frac{d^2+2dit-t^2}{4\alpha^{\frac{2}{mk}}x^2}\right)\frac{\mathrm{d}t}{t}+O\left(n^d\log x\right).$$

Applying the Stirling's approximation of $\Gamma(a+it)$, one has

$$\chi_{\mathbb{K}}(d+it)=\left(\frac{2\pi}{D^{1/m}t}\right)^{m\left(d+it-\frac{1}{2}\right)}e^{i\left(mt+\frac{\pi}{4}r_1\right)}\left(1+O\left(\frac{1}{t}\right)\right).$$

Thus

$$\mathcal{I}_n=\operatorname{Re}\frac{1}{\pi i}\int_1^{\infty}\left(x^{mk}n\right)^{d+it}\left(\frac{2\pi}{D^{1/m}t}\right)^{mk\left(d+it-\frac{1}{2}\right)}\exp\left(-\frac{t^2}{4\alpha^{\frac{2}{mk}}x^2}+ik\left(mt+\frac{\pi}{4}r_1\right)\right)\frac{\mathrm{d}t}{t}+O\left(n^dx^{\frac{mk}{2}-1+\epsilon}\alpha^{1/2}\right) \tag{2.4}$$

$$=\operatorname{Re}\frac{1}{\pi i}\int_1^{\infty}\left(x^{mk}n\right)^{d}\left(\frac{2\pi}{D^{1/m}t}\right)^{mk\left(d-\frac{1}{2}\right)}\exp\left(-\frac{t^2}{4\alpha^{\frac{2}{mk}}x^2}\right)$$
$$\times\exp\left(i\left(t\log\left(x^{mk}n\right)+mkt\log\left(\frac{2\pi}{D^{1/m}t}\right)+ik\left(mt+\frac{\pi}{4}r_1\right)\right)\right)\frac{\mathrm{d}t}{t}+O\left(n^dx^{\frac{mk}{2}-1+\epsilon}\alpha^{1/2}\right).$$

Now to compute the above integral, we use the method of stationary phase. The stationary point occurs at $t=2\pi x\left(n/D^k\right)^{\frac{1}{mk}}$. Let $y=t-2\pi x\left(n/D^k\right)^{\frac{1}{mk}}$ and $T=\left(x\left(n/D^k\right)^{\frac{1}{mk}}\right)^{3/5}$. We divide the

integral into two parts when $|y| \leq T$ and when $|y| > T$. In the first case, using a Taylor expansion, the above integral gives

$$\operatorname{Re}\ \frac{D^{1/m}}{2\pi^2 i}\left(xn^{\frac{1}{mk}}\right)^{\frac{mk}{2}-1}\int_{-T}^{T}\exp\left(-\pi^2\left(\frac{n}{D^k\alpha}\right)^{\frac{2}{mk}}+i\left(2\pi mkx\left(\frac{n}{D^k}\right)^{\frac{1}{mk}}+\frac{kr_1\pi}{4}-\frac{mk}{4\pi k}\left(\frac{D^k}{n}\right)^{\frac{1}{mk}}y^2\right)\right)$$
$$\times\left(1+O\left(\frac{1}{\left(x\left(n/D^k\right)^{\frac{1}{mk}}\right)^{1/5}}\right)\right)\mathrm{d}y. \tag{2.5}$$

It is well known that $\int_{-T}^{T} e^{-iz^2}\mathrm{d}z = e^{-\frac{i\pi}{4}} + O\left(1/T\right)$. Using this, (2.5) becomes

$$\frac{1}{\pi}\frac{D^{\frac{1}{2m}}}{\sqrt{mk}}\left(xn^{\frac{1}{mk}}\right)^{\frac{mk}{2}-1}\exp\left(-\pi^2\left(\frac{n}{D^k\alpha}\right)^{\frac{2}{mk}}\right)\left(\cos\left(2\pi mkx\left(\frac{n}{D^k}\right)^{\frac{1}{mk}}+\frac{\pi}{4}(kr_1-3)\right)+O\left(\left(x(n/D^k)^{\frac{1}{mk}}\right)^{-\frac{1}{10}}\right)\right). \tag{2.6}$$

Utilizing the exponential integral formula in (Chapter 4, [15]), for $1 \leq y \leq 2\pi x\left(n/D^k\right)^{\frac{1}{mk}} - T$,

$$\int_1^y \exp\left(i\left(t\log\left(x^{mk}n\right)+mkt\log\left(\frac{2\pi}{D^{1/m}t}\right)+mk\left(t+\frac{3\pi}{4}\right)\right)\right)\mathrm{d}t \ll \left(\log\left(\frac{1}{t}2\pi x\left(\frac{n}{D^k}\right)^{\frac{1}{mk}}\right)\right)^{-1}.$$

For the second case, we consider the range $|y| > T$. Using the upper bound established above, the contribution to the integral in (2.4) over the interval $1 \leq t \leq 2\pi x\left(n/D^k\right)^{\frac{1}{mk}} - T$ is

$$\ll \left(xn^{\frac{1}{mk}}\right)^{\frac{mk}{2}-\frac{3}{5}}\exp\left(-\left(\frac{n}{D^k\alpha}\right)^{\frac{2}{mk}}\right)+n^d x^{\frac{mk}{2}-1}\sqrt{\alpha}.$$

A similar bound also holds for the interval $t \geq 2\pi x\left(n/D^k\right)^{\frac{1}{mk}} + T$. Combining all these estimates together we obtain

$$\mathcal{I}_n = \frac{1}{\pi}\frac{D^{\frac{1}{2m}}}{\sqrt{mk}}\left(xn^{\frac{1}{mk}}\right)^{\frac{mk}{2}-1}\exp\left(-\pi^2\left(\frac{n}{D^k\alpha}\right)^{\frac{2}{mk}}\right)\left(\cos\left(2\pi mkx\left(\frac{n}{D^k}\right)^{\frac{1}{mk}}+\frac{\pi}{4}(kr_1-3)\right)\right)$$
$$+O\left(n^d x^{\frac{mk}{2}-1+\epsilon}\sqrt{\alpha}+\left(xn^{\frac{1}{mk}}\right)^{\frac{mk}{2}-\frac{3}{5}}\exp\left(-\left(\frac{n}{D^k\alpha}\right)^{\frac{2}{mk}}\right)\right).$$

Hence, the expression in (2.3) becomes

$$\frac{D^{\frac{1}{2m}}}{\pi\sqrt{mk}}x^{\frac{mk-1}{2}}\sum_{n=1}^{\infty}\frac{d_{\mathbb{K}}^{(k)}(n)}{n^{\frac{mk+1}{2km}}}\exp\left(-\pi^2\left(\frac{n}{D^k\alpha}\right)^{\frac{2}{mk}}\right)\cos\left(2\pi mkx\left(\frac{n}{D^k}\right)^{\frac{1}{mk}}+\frac{\pi}{4}(kr_1-3)\right)+O\left(x^{\frac{mk}{2}-\frac{3}{5}}\alpha^{\frac{1}{2}+\epsilon}\right).$$

Substituting this into (2.2), we complete the proof of our lemma. □

Before proceeding to the proof, we first establish the following lemma.

Lemma 2.3. *We have*

$$\max_{-1\leq h\leq 1}\Delta_{\mathbb{K}}^{(k)}\left(x^{mk}e^h\right) \geq \frac{\alpha^{\frac{1}{mk}}}{\sqrt{\pi}}\int_{-\infty}^{\infty}\Delta_{\mathbb{K}}^{(k)}\left(x^{mk}e^{u/x}\right)\exp\left(-u^2\alpha^{\frac{2}{mk}}\right)\mathrm{d}u + O(x^{mk-1}e^{-x^2}).$$

Proof. We start with

$$\begin{aligned}
&\frac{\alpha^{\frac{1}{mk}}}{\sqrt{\pi}}\int_{|u|\le x}\Delta_{\mathbb{K}}^{(k)}\left(x^{mk}e^{u/x}\right)\exp\left(-u^2\alpha^{\frac{2}{mk}}\right)\mathrm{d}u\\
&\le \max_{-1\le h\le 1}\Delta_{\mathbb{K}}^{(k)}\left(x^{mk}e^{h}\right)\frac{\alpha^{\frac{1}{mk}}}{\sqrt{\pi}}\int_{-\infty}^{\infty}\exp\left(-u^2\alpha^{\frac{2}{mk}}\right)\mathrm{d}u\\
&\le \max_{-1\le h\le 1}\Delta_{\mathbb{K}}^{(k)}\left(x^{mk}e^{h}\right).
\end{aligned}$$

Also, using the bound $\Delta_{\mathbb{K}}^{(k)}(y)=O(y)$, we find

$$\begin{aligned}
&\frac{\alpha^{\frac{1}{mk}}}{\sqrt{\pi}}\int_{x}^{\infty}\Delta_{\mathbb{K}}^{(k)}\left(x^{mk}e^{u/x}\right)\exp\left(-u^2\alpha^{\frac{2}{mk}}\right)\mathrm{d}u\\
&\ll x^{mk}\alpha^{\frac{1}{mk}}\int_{x}^{\infty}\exp\left(\frac{u}{x}-u^2\alpha^{\frac{2}{mk}}\right)\mathrm{d}u\\
&= x^{mk}\exp\left(\frac{1}{4x^2\alpha^{\frac{1}{mk}}}\right)\int_{x\alpha^{\frac{1}{mk}}}\exp\left(-\left(u-\frac{1}{2x\alpha^{\frac{1}{mk}}}\right)^2\right)\mathrm{d}u\\
&\ll x^{mk-1}e^{-x^2}.
\end{aligned}$$

A similar calculation gives

$$\frac{\alpha^{\frac{1}{mk}}}{\sqrt{\pi}}\int_{-\infty}^{-x}\Delta_{\mathbb{K}}^{(k)}\left(x^{mk}e^{u/x}\right)\exp\left(-u^2\alpha^{\frac{2}{mk}}\right)\mathrm{d}u\ll x^{mk-1}e^{-x^2}.$$

Combining all the above estimates, we conclude the proof. □

3. Proof of Theorem 1.1

Proof. We start the proof by applying the above lemma in Proposition 2.2 and truncate the infinite series in (2.1) at $n=X^{8/5}$. For the range $X/2\le x\le X^{8/5}$, we have

$$\begin{aligned}
\max_{-1\le h\le 1}\Delta_{\mathbb{K}}^{(k)}\left(x^{mk}e^{h}\right)\ge&\frac{D^{\frac{1}{2m}}}{\pi\sqrt{mk}}x^{\frac{mk-1}{2}}\sum_{n=1}^{X^{8/5}}\frac{d_{\mathbb{K}}^{(k)}(n)}{n^{\frac{mk+1}{2km}}}\exp\left(-\pi^2\left(\frac{n}{D^k\alpha}\right)^{\frac{2}{mk}}\right)\\
&\times\cos\left(2\pi mkx\left(\frac{n}{D^k}\right)^{\frac{1}{mk}}+\frac{\pi}{4}(kr_1-3)\right)+O\left(x^{\frac{mk}{2}-\frac{3}{5}}\alpha^{\frac{1}{2}+\epsilon}\right).
\end{aligned}\tag{3.1}$$

For $0\le\nu\le m$, P_ν denotes the set of all rational primes which are unramified in $\mathbb{K}$ and which split into exactly ν distinct $\mathcal{O}_{\mathbb{K}}$-prime ideals of degree 1. Let δ_ν be the Dirichlet density of P_ν as in the introduction and define $\mathcal{I}=\{1\le\nu\le m:\delta_\nu>0\}$. Then $R=|\mathcal{I}|$. We construct a resonator set $\mathcal{M}$ to be the set of all square free $n\in[C_1^{mk}\alpha,2^{mk}\alpha]$ such that n has $[\mu_\nu\log_2\alpha]$ distinct prime factors from P_ν for all $\nu\in I$ and no other, where C_1 is a constant satisfying $0<C_1<2$ and $\mu_\nu>0$ is a positive constant to be determined later. Let M be the cardinality of $\mathcal{M}$.

We notice that the value of $d_{\mathbb{K}}^{(k)}(n)$ is very large when $n\in\mathcal{M}$. Let $p\in P_\nu$ be prime. Then the factorization of $\mathcal{O}_{\mathbb{K}}$-prime ideal contains exactly ν prime ideals $\mathfrak{p}$ such that $N(\mathfrak{p})=p$. So that $d_{\mathbb{K}}^{(1)}(p)=\nu$. Also, for every prime ideal $\mathfrak{p}$, consider k tuples $(\mathfrak{n}_1,\mathfrak{n}_2,\cdots,\mathfrak{n}_k)$ such that $N(\mathfrak{n}_1\mathfrak{n}_2\cdots\mathfrak{n}_k)=p$ by choosing one of the $\mathfrak{n}_i=\mathfrak{p}$ and rest to be $\mathcal{O}_{\mathbb{K}}$. We see that $d_{\mathbb{K}}^{(k)}(p)=\nu k$. Similarly, one may get $d_{\mathbb{K}}^{(k)}(p^r)\ge\nu k$ for all $r\ge1$. We obtain that for $n\in\mathcal{M}$,

$$d_{\mathbb{K}}^{(k)}(n)\ge\prod_{\nu\in\mathcal{I}}(k\nu)^{[\mu_\nu\log_2\alpha]}.\tag{3.2}$$

We now proceed to apply Theorem 2.1. For this purpose, we consider

$$f(n)=\begin{cases} d_{\mathbb{K}}^{(k)}(n)n^{-\frac{mk+1}{2mk}}\exp\left(-\pi^2\left(\left(\frac{n}{D^k\alpha}\right)^{\frac{2}{mk}}\right)\right), & \text{if } n\leq X^{8/5},\\ 0, & \text{otherwise,}\end{cases}$$

$\lambda_n=2\pi mk(n/D^k)^{1/(mk)}$ and $\theta=(kr_1-3)\pi/4$. Also, we choose $A_1=8/5, A_2=3/2, A_3=1$ and $A_4=9/10$. Then using Theorem 2.1, there exists x satisfying $\sqrt{X}\leq x\leq 5X^{3/2}\log^2 X$ such that

$$\left|\sum_{n\leq X^{8/5}} f(n)\cos(\lambda_n x+\theta)\right|\geq \frac{\pi}{4e}\sum_{n\in\mathcal{M}} f(n)+O\left(X^{-1/2}e^{2M/C_1}\left(\sum_{\lambda_n\leq 4\alpha} a_n\right)\right)+O\left(\frac{X^{-9/10}}{\alpha}\sum_{n\leq X^{8/5}} a_n\right). \tag{3.3}$$

In order to estimate the lower bound of $\sum_{n\in\mathcal{M}} f(n)$, it is necessary to understand the size of $\mathcal{M}$. From the proof of Theorem 1 in [2], one find

$$M\asymp \frac{\alpha}{\log\alpha}\prod_{\nu\in\mathcal{I}}\frac{\left(\delta_\nu\log_2\alpha\right)^{[\mu_\nu\log_2\alpha]}}{[\mu_\nu\log_2\alpha]!}\asymp \alpha(\log\alpha)^{\kappa}(\log_2\alpha)^{-\mu},$$

where $\kappa=-1+\sum_{\nu\in\mathcal{I}}\mu_\nu(1+\log\delta_\nu-\log\mu_\nu)$ and $\mu=R/2$. We choose the parameter α such that M is of order $\log X$. In fact,

$$\alpha=C(k)(\log X)(\log_2 X)^{-\kappa}(\log_3 X)^{\mu},$$

where $C(k)$ is a small positive constant.

Note that $\log\alpha\asymp\log_2 X$ and $\log_2\alpha\asymp\log_3 X$. Also, for $n\in\mathcal{M}$, we have $(n/D^k\alpha)^{2/(mk)}\ll 1$. Using these and the lower bound in (3.2), we find that

$$\begin{aligned}\sum_{n\in\mathcal{M}} f(n)&\geq 2^{\frac{1}{2}(1-mk)}\alpha^{-\frac{mk+1}{2mk}}\prod_{\nu\in\mathcal{I}}(k\nu)^{[\mu_\nu\log_2\alpha]}\sum_{n\in\mathcal{M}}1\\ &\gg \alpha^{-\frac{mk+1}{2mk}}(\log\alpha)^{\sum_{\nu\in\mathcal{I}}\mu_\nu\log(k\nu)}M\\ &\gg \alpha^{\frac{mk-1}{2mk}}(\log\alpha)^{\kappa+\sum_{\nu\in\mathcal{I}}\mu_\nu\log(k\nu)}(\log_2\alpha)^{-\mu}\\ &\gg (\log X)^{\frac{mk-1}{2mk}}(\log_2 X)^{\frac{mk+1}{2mk}\kappa+\sum_{\nu\in\mathcal{I}}\mu_\nu\log(k\nu)}(\log_3 X)^{-\frac{mk+1}{2mk}\mu}\\ &\gg (\log X)^{\frac{mk-1}{2mk}}(\log_2 X)^{\beta}(\log_3 X)^{\gamma},\end{aligned} \tag{3.4}$$

where β and γ are defined as in Theorem 1.1. In the last step we optimize μ_ν in the power of $\log_2 X$ and choose $\mu_\nu=\delta_\nu(m\nu)^{2km/(km+1)}$. Our aim is to show that the error terms on the right hand side of (3.3) are smaller than the main term. Observe that $M\asymp C(k)\log X$. Again we choose $C(k)$ to be very small such that

$$e^{2M/C_1}\ll X^{1/4-\epsilon}.$$

Now let us compute the following sums:

$$\begin{aligned}\sum_{\lambda_n\leq 4\alpha} f(n)&\ll \sum_{n\leq 4\alpha} d_{\mathbb{K}}^{(k)}(n)n^{-\frac{mk+1}{2mk}}\\ &=\int_1^{4\alpha} t^{-\frac{mk+1}{2mk}}\,\mathrm{d}\left(\sum_{n\leq t} d_{\mathbb{K}}^{(k)}(n)\right)\\ &\ll \alpha^{\frac{mk-1}{2mk}}\log(4\alpha)^{k-1}\ll X^{\epsilon},\end{aligned} \tag{3.5}$$

where we used the bound $\sum_{n\le t} d_{\mathbb{K}}^{(k)}(n) \ll t(\log t)^{k-1}$. Similarly we find that

$$\sum_{n\le X^{8/5}} f(n) \ll X^{\frac{4(mk-1)}{5mk}} (\log X)^{k-1} \ll X^{\frac{4}{5}-\frac{1}{5mk}+\epsilon}. \tag{3.6}$$

The error can now be evaluated using the two estimates obtained above.

$$\begin{aligned}\text{Error} &\ll X^{-1/2} e^{2M/C_1} \left(\sum_{\lambda_n\le 4\alpha} f(n)\right) + \frac{X^{-9/10}}{\alpha}\left(\sum_{n\le X^{8/5}} f(n)\right)\\ &\ll X^{-1/4+\epsilon} + X^{-\frac{1}{10}-\frac{4}{5mk}+\epsilon} \ll X^{-\frac{1}{10}+\epsilon}.\end{aligned}$$

Employing (3.4) in (3.3) we conclude that for a large X,

$$\max_{\frac{X}{2}\le x\le 5X^{3/2}(\log X)^2} \left|\sum_{n\le X^{8/5}} f(n)\cos(\lambda_n x+\theta)\right| \gg (\log X)^{\frac{mk-1}{2mk}} (\log_2 X)^{\beta} (\log_3 X)^{\gamma}.$$

Hence, the inequality (3.1) implies

$$\max_{-1\le h\le 1} \Delta_{\mathbb{K}}^{(k)}\left(x^{mk}e^{h}\right) \gg x^{\frac{mk-1}{2}} (\log x)^{\frac{mk-1}{2mk}} (\log_2 x)^{\beta} (\log_3 x)^{\gamma}.$$

This completes the proof of our theorem. □

4. Some Special Examples

In this section, we give some examples to explain Theorem 1.1. In Examples 4.2 and 4.3, we look at normal and non-normal extensions, and these examples improve the ones given in [2]. We give an example of non-normal extension of degree 5 in Example 4.4. Similar computations for higher degree extensions can also be worked out (see Section 4 of [2]).

4.1. Setting $\mathbb{K} = \mathbb{Q}$ and choosing any $k \ge 2$, Theorem 1.1 reproduces the lower bound for the Piltz divisor problem previously proved by the second author [8].

4.2. We consider $\mathbb{K}$ to be a normal extension of $\mathbb{Q}$ with degree $m \ge 2$. Then the sets P_ν is empty for $1 \le \nu < m$. In this case, the Dirichlet densities $\delta_\nu = 0$ for $1 \le \nu < m$ and $\delta_m = 1/m$. Thus, for any integer $k \ge 1$, Theorem 1.1 gives

$$\Delta_{\mathbb{K}}^{(k)}(x) = \Omega\left((x\log x)^{\frac{mk-1}{2mk}} (\log_2 x)^{\frac{1}{2}k(mk+1)(mk)^{-2/(mk+1)}-\frac{1}{2}-\frac{1}{2mk}} (\log_3 x)^{-\frac{1}{4}-\frac{1}{4mk}}\right).$$

This result improves upon the bounds previously obtained by Hafner [4] and by Girstmair et al. [2].

4.3. We take $\mathbb{K}$ to be a non-normal extension of $\mathbb{Q}$ with $[\mathbb{K}:\mathbb{Q}] = 3$. Let us choose a field L such that $L = \mathbb{K}(\sqrt{D})$, where the discriminant D is not a perfect square. Then $\mathbb{K}^G$ is the minimal normal extension of $\mathbb{Q}$ and its Galois group G is isomorphic to S_3, the symmetric group of three elements. Also, L is a quadratic extension of $\mathbb{K}$. So, the Galois group $H = \mathrm{Gal}(L/K)$ is isomorphic to the cyclic group $\mathbb{Z}_2$. Then the Dirichlet densities are given by $\delta_1 = 1/2$, $\delta_2 = 0$ and $\delta_3 = 1/6$. Then for any $k \ge 1$,

$$\Delta_{\mathbb{K}}^{(k)} = \Omega\left((x\log x)^{\frac{1}{2}-\frac{1}{6k}} (\log_2 x)^{\beta} (\log_3 x)^{-\frac{1}{2}-\frac{1}{6k}}\right),$$

where

$$\beta = \frac{(3k+1)}{12} k^{(3k-1)/(3k+1)} \left(3^{\frac{3k-1}{3k+1}}+1\right) - \frac{1}{2} - \frac{1}{6k}.$$

4.4. Let $f(T) = T^5+20T+16$. Then $f(T)$ is an irreducible polynomial over $\mathbb{Q}$. We consider $\mathbb{K} = \mathbb{Q}(\xi)$, where ξ is a root of f and $m = [\mathbb{K} : \mathbb{Q}] = 5$. It is clear that K is not normal extension. We denote $\mathbb{K}^G$ as the Galois closure of the extension $\mathbb{K}/\mathbb{Q}$. Then $G = \mathrm{Gal}(\mathbb{K}^G/\mathbb{Q}) \cong A_5$ and $H = \mathrm{Gal}(\mathbb{K}^G/\mathbb{K}) \cong A_4$, where A_n the alternating group on n elements. There are exactly three non-empty sets P_ν ($\nu = 1, 2, 5$), which are in $\mathbb{K}$. The corresponding non-zero Dirichlet densities are $\delta_1 = 1/4$, $\delta_2 = 1/3$ and $\delta_5 = 1/60$. Thus, for each $k \geq 2$,

$$\Delta_{\mathbb{K}}^k(x) = \Omega\left((x\log x)^{\frac{1}{2}-\frac{1}{10k}}(\log_2 x)^{\beta}(\log_3 x)^{-\frac{3}{4}-\frac{3}{20k}}\right),$$

where

$$\beta = \frac{(5k+1)}{600}k^{(5k-1)/(5k+1)}\left(15+5\cdot 2^{\frac{15k+1}{5k+1}}+5^{\frac{10k}{5k+1}}\right)-\frac{1}{2}-\frac{1}{10k}.$$

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